Md Halim Mondol

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RESEARCH INTERESTS

Large Language Models (LLMs), Generative AI, Deep Learning (DL), Power Electronics

EDUCATIONS

- **North Carolina State University**, Raleigh, NC
  – Ph.D. in Electrical Engineering (CGPA: 3.81/4.00) August 2022 - June 2027 (Expected)
  Advisor: Edgar Lobaton
  – MS in Electrical Engineering (CGPA: 3.80/4.00) August 2022 - May 2024
- **Rajshahi University of Engineering and Technology**, Dhaka, Bangladesh
  – B.Sc. in Electronics and Telecommunication Engineering (CGPA: 3.68/4.00) January 2016 - March 2021

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, HTML, Assembly, Bash, Shell
Frameworks/Libraries: PyTorch, FAISS, Scikit-learn, Pandas, Kaggle, LangChain, Hugging Face Transformers
AI/ML Domains: Machine Learning, Large Language Models, Natural Language Processing, Generative AI, Transformer, AI Model Fine-tuning, Retrieval-Augmented Generation, Prompt Engineering
Cloud/Tools: GCP, AWS SageMaker, AppRunner, ECR, S3, Docker, Git/GitHub, Streamlit, SLURM, Linux, FastAPI

WORK EXPERIENCES

- **Graduate Research Assistant**, ARoS Lab, NC State University Raleigh, NC [May 2024 – Present]
  – Developed end-to-end ML systems, including generative models (LTVAE) and scientific RAG applications.
  – Trained and fine-tuned deep-learning models using PyTorch, FAISS, RDKit, and GPU workflows on AWS/GCP.
  – Built LLM applications with Streamlit, conversation memory, and guardrails for safe, reliable outputs.
  – Containerized and deployed ML services using Docker and FastAPI for scalable experimentation.
  – Applied ML to molecular modeling and other scientific data analysis tasks.
- **Graduate Teaching Assistant**, NC State University Raleigh, NC [Jan 2024 – May 2024]
  – Teaching assistant for Deep Learning course: graded neural network homework, managed GitHub Classrooms, and set up/monitored an NSF-supported JupyterHub server.
  – Installed PyTorch Docker images, prepared student environments, and troubleshot server-side errors.
- **Electrification Research Intern**, ABB Corporate Research Raleigh, NC [May 2023 – Aug 2023]
  – Designed energy harvesting circuits using current transformers for onboard measurement systems.
  – Developed real-time motor health monitoring system with DAQ device and PCB design in Altium.

PROJECTS

- **Phosphorus Knowledge Hub:** Built the upstream workflow for a scientific RAG system, covering synthetic data generation, automated report creation, formula/range extraction, and metadata indexing. Added guardrails and physics-based validation tools to ensure safe, accurate, and physically consistent LLM outputs.
- **AWS Deployment of LTVAE Model:** Designed and deployed an LTVAE-based molecular inference API (SMILES → latent vectors → property predictions) using FastAPI and PyTorch. Containerized the full pipeline with Docker and launched a scalable cloud service through AWS ECR and App Runner, including RDKit validation, property-head MLP modules, and real-time REST endpoints.
- **LTVAE: Molecular Generation and Property Prediction:** A hybrid deep-learning framework was developed using a Bi-LSTM encoder, variational latent space, and Transformer decoder to learn continuous molecular representations from SMILES strings. High reconstruction quality was achieved, with 98.6% token-level accuracy, 97.4% chemical validity, and strong generalization to structurally distinct dye molecules.

PUBLICATIONS

– **Mondol, M.H.**, Wei, Q., Lobaton, E., "LTVAE: Property-Aware Latent Representations from SMILES for VOC–Dye Screening and Molecular Design", *Under Review*, *J. Chem. Info. Model.*, 2025.
– Jamalzadegan, S., Penumudy, A., Mativenga, B., **Mondol, M.H.**, et al., "AI-Powered Colorimetric Sensing with LLMs", *AIChE Annual Meeting*, 2025.